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A Note on

A toy model of credit frictions

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1. Economic question and literature

Why can a modest deterioration in financing conditions cause a large fall in investment and employment? When borrowing is secured by productive assets, current investment affects tomorrow's collateral while the expected resale price of that collateral affects credit available today. A decline in asset values can therefore tighten borrowing precisely when firms would otherwise smooth their expenditure. The interaction is an amplification mechanism, not an assumption that every adverse shock must generate a recession.

Kiyotaki and Moore [1997] establish this interaction between credit limits and asset prices. Iacoviello [2015] studies financial business cycles with constrained borrowers and financial intermediaries. The present model isolates the entrepreneur's collateral and working-capital margins. Households save, entrepreneurs borrow and the wage bill competes with capital for scarce financing. This makes the model useful for asking whether movements in credit, investment and employment are consistent with a collateral disturbance rather than a productivity disturbance.

Gerali et al. [2010] estimate a richer euro-area model with banks, lending spreads and bank capital. Their contribution helps distinguish a borrowing constraint from a banking-sector transmission mechanism. Here intermediaries are competitive, so a deposit–loan spread is not a separate observable implied by the model. Investment adjustment costs follow the mechanism used by Christiano et al. [2005]: changing investment rapidly is costly, which makes the timing of investment and the value of installed capital important. The exercise preserves these mechanisms while leaving empirical identification and estimation to the student.

2. The model

All interest rates below are gross real returns. Deposits and loans chosen at t repay at $t+1$. Installed capital K_{t-1} produces at t, while K_t is chosen at the end of t. Conditional expectations use information at t.

2.1 Households

Patient households have discount factor β_H and consumption curvature $\sigma_C > 0$. Define external habits $X_t^H = h_c \overline{C}_{t-1}^H$, where the bar in this definition denotes the aggregate consumption taken as given by an individual household. In symmetric equilibrium $X_t^H = h_c C_{t-1}^H$. The household's objective is

$$\max_{\{C_{t+\tau}^H, d_{t+\tau}, H_{t+\tau}\}_{\tau \geq 0}} E_t \sum_{\tau=0}^{\infty} \beta_H^\tau \left[\varepsilon_{t+\tau}^C \frac{(C_{t+\tau}^H - X_{t+\tau}^H)^{1-\sigma_C}}{1-\sigma_C} - \chi \frac{H_{t+\tau}^{1+\sigma_H}}{1+\sigma_H} \right]. \tag{1}$$

The logarithmic limit applies when $\sigma_C = 1$. The habit stock is external, so the individual does not differentiate future aggregate habits. The budget is

$$C_t^H + d_t = R_{t-1} d_{t-1} + H_t w_t - T_t. \tag{2}$$

Here d_t is a real deposit, w_t a real wage and T_t a lump-sum tax. Writing the budget residual as $\mathcal{B}_t^H$ (income minus expenditure), the Lagrangian is

$$\mathcal{L}_t^H = E_t \sum_{\tau \geq 0} \beta_H^\tau \left[\varepsilon_{t+\tau}^C \frac{(C_{t+\tau}^H - X_{t+\tau}^H)^{1-\sigma_C}}{1-\sigma_C} - \chi \frac{H_{t+\tau}^{1+\sigma_H}}{1+\sigma_H} + \lambda_{t+\tau}^H \mathcal{B}_{t+\tau}^H \right].$$

The first-order conditions are

$$\lambda_t^H = \varepsilon_t^C (C_t^H - X_t^H)^{-\sigma_C}, \tag{3}$$

$$\lambda_t^H = \beta_H R_t E_t \lambda_{t+1}^H, \tag{4}$$

$$\chi H_t^{\sigma_H} = \lambda_t^H w_t. \tag{5}$$

The Euler equation prices saving. The last equation equates the marginal cost of work to the consumption value of its wage.

2.2 Entrepreneurs

Entrepreneurs are less patient, $\beta_E < \beta_H$. Their borrowing capacity is

$$l_t \leq \varepsilon_t^M m_K E_t(q_{t+1}) K_t / R_t - m_H h_t w_t. \tag{6}$$

The baseline is a local approximation around a binding constraint and therefore uses equality. The parameter m_K is the pledgeable fraction of capital value and m_H measures the wage bill that uses up financing capacity. The positive shock ε_t^M changes pledgeability. A lower expected q_{t+1} reduces borrowing capacity even before the installed capital stock changes.

Entrepreneurs maximize

$$\max_{\{C_{t+\tau}^E, K_{t+\tau}, I_{t+\tau}, l_{t+\tau}, h_{t+\tau}\}_{\tau \geq 0}} E_t \sum_{\tau=0}^{\infty} \beta_E^\tau \varepsilon_{t+\tau}^C \frac{(C_{t+\tau}^E - X_{t+\tau}^E)^{1-\sigma_C}}{1-\sigma_C}, \tag{7}$$

where $X_t^E = h_c \overline{C}_{t-1}^E$ is also external. Production, the balance sheet and capital accumulation are

$$Y_t = \varepsilon_t^A K_{t-1}^{\alpha} h_t^{1-\alpha}, \tag{8}$$

$$C_t^E + I_t + w_t h_t + R_{t-1} l_{t-1} = Y_t + l_t, \tag{9}$$

$$I_t \varepsilon_t^I [1 - S(I_t/I_{t-1})] = K_t - (1-\delta) K_{t-1}, \tag{10}$$

with $S(v) = \kappa(v-1)^2/2$. Thus $S(1) = S'(1) = 0$. The steady-state investment ratio is one, not an additional parameter. Define $\mathcal{B}_t^E$ as the balance-sheet residual, $\mathcal{C}_t^E = \varepsilon_t^M m_K E_t(q_{t+1}) K_t / R_t - m_H h_t w_t - l_t$ and $\mathcal{K}_t = I_t \varepsilon_t^I [1 - S(I_t/I_{t-1})] + (1-\delta) K_{t-1} - K_t$. The Lagrangian preserves all three constraints:

$$\mathcal{L}_t^E = E_t \sum_{\tau \geq 0} \beta_E^{\tau} \left[\varepsilon_{t+\tau}^C \frac{(C_{t+\tau}^E - X_{t+\tau}^E)^{1-\sigma_C}}{1-\sigma_C} + \lambda_{t+\tau}^E \mathcal{B}_{t+\tau}^E + \lambda_{t+\tau}^E \varphi_{t+\tau}^E \mathcal{C}_{t+\tau}^E + \lambda_{t+\tau}^E q_{t+\tau} \mathcal{K}_{t+\tau} \right].$$

Both q_t and φ_t^E are measured in consumption units by scaling their multipliers with λ_t^E. Let $v_t = I_t / I_{t-1}$. The first-order conditions, before dividing by marginal utility, are

$$C_t^E: \quad \varepsilon_t^C (C_t^E - X_t^E)^{-\sigma_C} - \lambda_t^E = 0, \tag{11}$$

$$l_t: \quad \lambda_t^E - \beta_E R_t E_t \lambda_{t+1}^E - \lambda_t^E \varphi_t^E = 0, \tag{12}$$

$$I_t: \quad -\lambda_t^E + \lambda_t^E q_t \varepsilon_t^I [1 - S(v_t) - v_t S'(v_t)] + \beta_E E_t [\lambda_{t+1}^E q_{t+1} \varepsilon_{t+1}^I S'(v_{t+1}) v_{t+1}^2] = 0, \tag{13}$$

$$K_t: \quad \beta_E E_t \Big[\lambda_{t+1}^E \left((1-\delta) q_{t+1} + \alpha Y_{t+1}/K_t \right) \Big] - \lambda_t^E q_t + \lambda_t^E \varphi_t^E \varepsilon_t^M m_K E_t(q_{t+1})/R_t = 0, \tag{14}$$

$$h_t: \quad \lambda_t^E [(1-\alpha) Y_t/h_t - w_t] - \lambda_t^E \varphi_t^E m_H w_t = 0. \tag{15}$$

Equivalently, the normalized equations used for the simulation are

$$\lambda_t^E = \varepsilon_t^C (C_t^E - X_t^E)^{-\sigma_C}, \tag{16}$$

$$1 - \varphi_t^E = \beta_E R_t E_t (\lambda_{t+1}^E / \lambda_t^E), \tag{17}$$

$$q_t \varepsilon_t^I = 1 + q_t \varepsilon_t^I \frac{\kappa}{2} [1 + (3v_t - 4) v_t] + \beta_E E_t [(\lambda_{t+1}^E/\lambda_t^E) q_{t+1} \varepsilon_{t+1}^I \kappa (1 - v_{t+1}) v_{t+1}^2], \tag{18}$$

$$q_t = \beta_E E_t [(\lambda_{t+1}^E/\lambda_t^E)((1-\delta) q_{t+1} + \alpha Y_{t+1}/K_t)] + \varphi_t^E \varepsilon_t^M m_K E_t(q_{t+1})/R_t, \tag{19}$$

$$w_t = \frac{(1-\alpha) Y_t}{h_t (1 + m_H \varphi_t^E)}. \tag{20}$$

The expected return on capital remains inside the stochastic discount-factor expectation. Factoring it out would discard a covariance and is not an exact nonlinear identity. A larger multiplier raises the financing wedge on labor and capital. It does not change households' preference parameter β_H.

2.3 Financial intermediaries

Competitive intermediaries collect deposits and lend to entrepreneurs. If b_t denotes their holdings of public bonds, their portfolio satisfies $d_t = l_t + b_t$. The expected one-period payoff is

$$\Pi_{t+1}^B = R_t^E l_t + R_t b_t - R_t d_t. \tag{21}$$

Competition implies $R_t^E = R_t$. The baseline has zero public debt and taxes finance public purchases, so $d_t = l_t$. Banks do not add a separate spread.

2.4 Equilibrium conditions

Labor clears at $H_t = h_t$, and the goods market satisfies

$$C_t = C_t^E + C_t^H, \tag{22}$$

$$Y_t = C_t + I_t + G_t, \qquad G_t = g^y \bar{Y} \varepsilon_t^G. \tag{23}$$

For $s \in \{A, G, C, M, I\}$, shocks follow $\log \varepsilon_t^s = \rho_s \log \varepsilon_{t-1}^s + \eta_t^s$. Innovations have zero mean and specified standard deviations.

2.5 The friction in steady state

The deposit Euler equation gives $\bar{R} = 1/\beta_H$. The loan condition gives

$$\bar{\varphi}^E = 1 - \beta_E \bar{R} = 1 - \beta_E/\beta_H > 0.$$

This establishes that the constraint binds at the deterministic steady state. It justifies a local binding-regime approximation, not a claim that the inequality must bind after every possible shock. Positive consumption net of habits and a feasible loan stock must also be checked.

3. Simulation and empirical starting point

The accompanying `credit_RBC.mod` is nonlinear in levels. It uses $\beta_H = .993$, $\beta_E = .980$, $\delta = .025$, $\alpha = .30$, $g^y = .2$, $h_c = .3$, $\sigma_C = \sigma_H = 1$, $m_K = .8$, $m_H = 1$ and $\kappa = 4$. Hours are normalized to 1/3 in steady state and χ is recalibrated accordingly. All five shock persistence parameters are .95. The supplied unit standard deviations produce normalized first-order responses, not estimated shock scales or a large-shock nonlinear experiment.

Run the supplied simulation and inspect residuals and determinacy before constructing observables. Output, investment, credit stocks and a real lending rate have different units and timing. The downloader supplies only raw output and investment responses. Output is a chained-volume series, whereas the investment series is at current prices. An investment deflator or an appropriate volume series is therefore required before treating both as real model quantities. These archived series end in 2023Q4. They are not a corporate-credit measure or an estimation-ready data set. Choose and document additional series, deflators, aggregation, transformations and measurement equations as part of the assessment. No estimation settings, posterior results or corrected data are supplied here.

References

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